

Functional Annotation Report: PP_3303 (Q88HQ0) — β -Ketoacyl-[ACP] Synthase II (KAS II / FabF)

Organism: *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440), PSEPK **Gene / locus:** PP_3303 · **UniProt:** Q88HQ0 · **Length:** 437 aa · **EC:** 2.3.1.179 **Family:** Thiolase-like superfamily; β -ketoacyl-ACP synthase I/II (KAS). **Domains:** Beta-ketoacyl_synthase (IPR000794), Ketoacyl_synth_N (IPR014030), Ketoacyl_synth_C (IPR014031), 3-oxoacyl-ACP_synth-2 (IPR017568), Ketoacyl_synth_AS (IPR018201).

1. Summary (Answer to the Research Question)

PP_3303 encodes **β -ketoacyl-[acyl-carrier-protein] synthase II (KAS II, the FabF condensing enzyme; EC 2.3.1.179)**, a soluble **cytoplasmic** enzyme of the dissociated **type II fatty acid synthase (FAS-II)**. Its primary function is to catalyze the **carbon-carbon bond-forming, decarboxylative Claisen condensation** that elongates a growing fatty acyl chain by two carbons in each turn of the FAS-II elongation cycle: it transfers the acyl group from **acyl-ACP** onto an active-site cysteine and condenses it with a **malonyl-ACP** extender unit to yield a **3-oxoacyl-ACP** (β -ketoacyl-ACP), releasing holo-ACP and CO₂. Its diagnostic, curated substrate reaction is the elongation of **palmitoleoyl-ACP (C16:1 cis-9)** to the **β -keto precursor of cis-vaccenoyl-ACP (C18:1 cis-11)**, identifying it as the KAS II isoform responsible for **cis-vaccenic acid** production and hence for the **temperature-dependent tuning of membrane fatty-acid (homeoviscous) composition**.

2. Verified Identity

The target was cross-checked against the supplied UniProt record and confirmed unambiguous:

Attribute	Value (confirmed)
Protein name	3-oxoacyl-[ACP] synthase 2 (KAS II / FabF)
EC number	2.3.1.179 (KAS II) — distinct from EC 2.3.1.41 (KAS I/FabB) and EC 2.3.1.180 (KAS III/FabH)
Length	437 residues
N-terminal thiolase motif	<code>. . . IVVTGTGVVSPLGC . . .</code> (conserved KAS Cys-cap region)
Curated reaction (specific)	(9Z)-hexadecenoyl-[ACP] + malonyl-[ACP] + H ⁺ = 3-oxo-(11Z)-octadecenoyl-[ACP] + holo-[ACP] + CO ₂
Curated reaction (general)	acyl-[ACP] + malonyl-[ACP] + H ⁺ = 3-oxoacyl-[ACP] + holo-[ACP] + CO ₂
Pathway	Lipid metabolism; fatty acid biosynthesis

Because the annotated reaction and EC number are the **hallmark KAS II activity** (elongation of long, including C16:1, acyl-ACP; production of the cis-vaccenate precursor), there is no gene-symbol ambiguity: this is genuinely a bacterial FabF/KAS II from *P. putida* KT2440.

Sequence-level confirmation of the catalytic apparatus (this work): Direct inspection of the 437-aa sequence recovers all diagnostic elongating-KAS motifs — the N-terminal thiolase signature **TGTGVVSPLGC** (res. 20–30), the nucleophilic **Cys184** in the acyl-cysteine motif `APVTAC184AAGV`, the first catalytic histidine **His325** in the `NAH325ATST` motif, and the second catalytic histidine **His361** in the strictly conserved elongation signature `KSATGH361LLGAAG` (KSxxGHxxG). This reconstitutes the canonical **Cys-His-His triad** (cf. *E. coli* FabF Cys163/His303/His340), giving direct sequence evidence — not merely orthology — that PP_3303 is a catalytically competent KAS II.

Phylogenetic placement (this work): Global pairwise-identity comparison places PP_3303 firmly in the **FabF/KAS II clade**: it is most similar to *E. coli* FabF/KAS II (48.7%) and to the KT2440 housekeeping fabF PP_1916 (49.8%), clearly less similar to KAS I/FabB (*E. coli* 41.7%; KT2440 PP_4175 39.7%), and least similar to KAS III/FabH (29.1%). This independently corroborates the EC 2.3.1.179 assignment and **excludes** a FabB (KAS I) or FabH (KAS III) identity. The ~50% identity to PP_1916 indicates the two are anciently diverged KAS-II paralogs rather than recent duplicates.

2a. Paralog census and genomic context (important caveat)

A UniProt census of *P. putida* KT2440 FAS-II condensing enzymes shows PP_3303 is **one of several KAS-family paralogs**:

Locus	UniProt	Annotation	Note
PP_4175	Q88FC3	FabB / KAS I (EC 2.3.1.41)	unsaturated FA initiation
PP_1916	Q88LL4	FabF / KAS II (EC 2.3.1.179)	housekeeping fabF, in fab cluster beside fabG (PP_1914)
PP_3303	Q88HQ0	KAS II (EC 2.3.1.179)	this study; standalone paralog
PP_2780	Q88J69	KAS-II-like	accessory paralog
PP_2778	Q88J71	KAS-II-like	accessory paralog

Critically, PP_3303 is **not** located in the core fatty-acid biosynthesis operon. Its genomic neighborhood (PP_3300–PP_3305) is an **efflux/stress-response island**: a TetR-family transcriptional regulator (PP_3300), RND efflux transporter + membrane-fusion components (PP_3301/PP_3302), a Bcr/CflA-family efflux transporter (PP_3304), and a TerC-family membrane protein (PP_3305). The canonical housekeeping elongation synthase is instead **PP_1916 (fabF)**, co-clustered with **fabG (PP_1914)**. This indicates that while PP_3303's **molecular function is unambiguously KAS II condensation**, its **in vivo physiological deployment may be accessory or condition/stress-responsive** rather than the primary housekeeping elongation role — a distinction that should temper direct transfer of *E. coli* fabF phenotypes to PP_3303.

3. Primary Function — The Reaction Catalyzed

FAS-II synthesizes fatty acids on the small acyl carrier protein (ACP) through iterative two-carbon elongation cycles (condensation → ketoreduction → dehydration → enoyl reduction). PP_3303 performs the **committed condensation (C–C bond-forming) step** of the cycle [PMID 39175097]. KAS enzymes catalyze a **decarboxylative Claisen condensation** by a **two-step ping-pong mechanism**: (1) the acyl donor acylates the active-site cysteine (acyl-enzyme intermediate); (2) malonyl-ACP is decarboxylated to a carbanion/enolate that attacks the acyl-

enzyme, forming the new C–C bond and yielding β -ketoacyl-ACP [PMID 22017312]. The **elongating** synthases KAS I (FabB) and KAS II (FabF) use an **acyl-ACP** donor (in contrast to the priming KAS III/FabH, which uses acyl-CoA) [PMID 22017312].

4. Substrate Specificity

The curated Q88HQ0 reaction — **palmitoleoyl-ACP (C16:1 Δ 9) + malonyl-ACP $\rightarrow$ 3-oxo-cis-vaccenoyl-ACP (C18:1 Δ 11) + CO₂** — is the defining KAS II activity. Biochemically, **FabF/KAS II efficiently elongates saturated C14 and unsaturated C16:1 acyl-ACP substrates**, whereas the paralogous **KAS I (FabB) has reduced activity beyond C12** [PMID 36048156]. This chain-length preference places PP_3303 at the **terminal, long-chain elongation steps** of FAS-II and makes it the enzyme that produces **cis-vaccenic acid (C18:1 cis-11)**, the principal long-chain monounsaturated fatty acid of enterobacterial and pseudomonad membranes. Recent structural work shows FabF achieves substrate selectivity through hydrophobic cavities and "front/middle/back door" gating residues that stabilize specific acyl chain lengths [PMID 39175097], with mobile active-site loops enforcing an ordered "gating" mechanism [PMID 32265440].

5. Catalytic Mechanism & Active Site

PP_3303 adopts the **thiolase fold** and employs a conserved **Cys-His-His catalytic triad** [PMID 11286890]. Detailed mutagenesis of a KAS II (*S. pneumoniae* FabF) assigns the roles [PMID 16618705]: - **Nucleophilic active-site cysteine** — required for acyl-enzyme formation and overall condensation (UniProt annotates the KAS active-site residue on Q88HQ0). - **Two histidines with distinct electronic states** — one (His337 in Sp numbering) stabilizes the malonyl thioester oxyanion in the transition state; the other (His303) deprotonates a structured active-site water to drive decarboxylation and bicarbonate release. - **A conserved lysine** controls the histidine electronic state; a **gatekeeper phenylalanine (Phe396)** enforces ordered substrate addition.

Quantum-chemical analysis of *E. coli* FabF corroborates that conserved **Lys and His residues facilitate proton transfer during C–C bond formation** [PMID 39175097]. The catalytic cysteine's reactivity is directly demonstrated by the antibiotic **cerulenin**, which is **covalently**

bonded to the active-site cysteine [PMID 10037680]; FabF is likewise the target of platensimycin-class antibacterials [PMID 18336284], underscoring both the mechanism and druggability of this enzyme class.

6. Subcellular Localization

As a component of the **soluble, dissociated FAS-II system**, PP_3303 operates in the **cytoplasm**, acting on substrates tethered to the phosphopantetheine arm of ACP. Its β -ketoacyl-ACP products are processed by the downstream cytosolic tailoring enzymes (FabG, FabA/FabZ, FabI) and ultimately channelled into **membrane phospholipid** biosynthesis at the inner membrane. There is no signal peptide or membrane-spanning domain in the sequence, consistent with a cytoplasmic assignment.

7. Pathway Context & Physiological Role

Pathway: Bacterial type II fatty acid biosynthesis (FAS-II) → membrane phospholipid acyl chains. PP_3303 is the elongation condensing enzyme that governs production of long-chain (C18) unsaturated fatty acids.

Homeoviscous / thermal regulation: Classic *E. coli* genetics established that **fabF-null mutants lack both cis-vaccenic acid synthesis and thermal regulation**, and that **synthase II (FabF) is "the sole enzyme regulating the temperature-dependent composition of the membrane phospholipid acyl chains"** [PMID 6337151]. At lower temperatures FabF activity increases the proportion of the longer C18:1 (cis-vaccenate) chain, maintaining membrane fluidity — a role directly attributable to PP_3303's substrate preference.

Relevance to *P. putida* physiology: In the solvent-tolerant *P. putida* lineage, the very unsaturated fatty acid produced by FabF (**cis-vaccenate**) is the substrate for the periplasmic **cis/trans isomerase (Cti)**, which converts cis- to trans-unsaturated fatty acids as the principal **short-term adaptation to organic solvents**, complemented by longer-term cardiolipin changes [PMID 17564601]. Adaptive remodeling of membrane fatty-acid and phospholipid composition is a well-documented solvent-tolerance mechanism in *P. putida* and related pseudomonads [PMID 12142492,

P 11030576

Thus PP_3303 supplies the biosynthetic precursor pool that underlies both **temperature** and **solvent-stress** membrane adaptation in this biotechnologically important organism.

8. Supported vs. Refuted Hypotheses

Hypothesis	Status	Basis
PP_3303 is a KAS II/FabF condensing enzyme (EC 2.3.1.179)	Supported	UniProt curated EC/reaction; domain architecture [PMID 22017312, 39175097]
It elongates palmitoleoyl-ACP → cis-vaccenate precursor (long-chain specificity)	Supported	Curated reaction; FabF C14/C16:1 preference [PMID 36048156]
Uses Cys-His-His triad, ping-pong Claisen condensation	Supported	[PMID 16618705, 11286890, 39175097, 10037680]
Cytoplasmic FAS-II enzyme	Supported	Type II FAS biology; no signal/TM segment
Governs temperature-dependent membrane fatty-acid composition	Supported (by orthology)	<i>E. coli</i> fabF genetics [PMID 6337151]
It is KAS I (FabB, EC 2.3.1.41) or KAS III (FabH)	Refuted	EC 2.3.1.179 and the C16:1→C18:1 reaction are KAS II-specific
Catalytic Cys-His-His triad intact in the actual Q88HQ0 sequence	Supported	Cys184/His325/His361 + KSATGH & thiolase motifs (this work)
PP_3303 is the sole/housekeeping fabF of KT2440	Refuted / qualified	Housekeeping fabF is PP_1916 (in fabG cluster); PP_3303 is a standalone paralog in an efflux/stress island (this work)
PP_3303 belongs to KAS II (FabF) clade, not KAS I/III	Supported	Global identity: 48.7% EcFabF, 41.7% EcFabB, 29.1% EcFabH (this work)

9. Limitations & Future Directions

- **Direct biochemical or genetic characterization of PP_3303 specifically in *P. putida* KT2440 is not available in the literature**; functional assignment rests on the curated UniProt reaction, direct sequence verification of the catalytic triad, and strong orthology to extensively characterized *E. coli*, *S. pneumoniae*, and *Synechocystis* KAS II enzymes. The catalytic identity is robust; the *in vivo* physiological role is inference-by-homology.
 - **PP_3303 is a paralog, not the housekeeping fabF**. KT2440 encodes a dedicated fabF (PP_1916, in the fabG cluster), a FabB/KAS I (PP_4175), and two further KAS-II-like genes (PP_2778/PP_2780). PP_3303 sits in an efflux/stress genomic island. The **division of labor, relative flux, expression conditions, and possible functional redundancy** among these paralogs have not been experimentally dissected, so the cis-vaccenate/thermal-regulation role (established for the single *E. coli* fabF) may in KT2440 be carried primarily by PP_1916, with PP_3303 accessory or condition-specific.
 - Future work: paralog-resolved knockouts (Δ PP_3303 vs Δ PP_1916, and double mutants) with temperature/solvent-shift lipidomics; RNA-seq to define the condition-dependence of PP_3303 (given its efflux-island context and adjacent TetR regulator); enzyme kinetics on defined acyl-ACP substrates; structural determination to confirm the gating residues.
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References (PMIDs)

- 39175097 — FabF catalyzes C–C bond formation; bimodal specificity; catalytic Lys/His (2024)
- 36048156 — FabB/FabF substrate selectivity; FabF elongates C14/C16:1 (2022)
- 32265440 — Gating mechanism of elongating β -ketoacyl-ACP synthases (2020)
- 22017312 — Substrate recognition by β -ketoacyl-ACP synthases; ping-pong/Claisen (2011)
- 16618705 — Active-site residues (Cys/His/His/Lys/Phe) of KAS II FabF mechanism (2006)
- 11286890 — KAS I structure; Cys-His-His triad shared by KAS II (2001)
- 11152607 — KAS II crystal structure (*Synechocystis*), condensing-enzyme classification (2001)
- 10037680 — Cerulenin covalently bound to active-site cysteine of KAS (1999)
- 18336284 — Platensimycin targets FabF/FabB (2008)

- 6337151 — FabF/synthase II is the sole thermal regulator of membrane acyl-chain composition (1983)
- 17564601 — cis/trans isomerase and cardiolipin in *P. putida* membrane adaptation (2007)
- 12142492 — Mechanisms of solvent tolerance in Gram-negative bacteria (2002)
- 11030576 — Phospholipid compositional changes in pseudomonads with toluene (2000)

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